# Efficiency−TFP Analysis

## Austria

*MAPS Project — Deliverable 4.2 (WP4: Improving Integrated Assessment Modelling)*

Period: 1960–2019 • Observations: 60 • Date: 22 May 2026

This report documents the Efficiency−TFP cointegration analysis for Austria (1960–2019). Following De Ketelaere et al. (2026) and Santos et al. (2021), it tests whether changes in Total Factor Productivity (TFP) are explained by improvements in final−to−useful exergy efficiency, and assesses the stability of useful exergy intensity for scenario analysis.

### Steps of the Analysis

1. Output elasticities (a_K, a_L) from income data
2. Labour inputs: unadjusted and quality−adjusted
3. TFP: unadjusted and quality−adjusted (z−normalised to base year)
4. Exergy efficiency and useful exergy intensity
5. Unit root tests (ADF and PP) on TFP and efficiency series
6. VAR estimation and residual diagnostics
7. Johansen cointegration tests: Hz (no deterministic) and Hc (restricted constant; no trend)
8. Useful exergy intensity constancy assessment

### Key Results

**i) Long−run TFP−Efficiency relationship [Hc specification — restricted constant; no trend]:**

**$TFP = \exp(1.1214) \times EFF^{0.3627}$ (quality−adjusted TFP)**

(z−form: $z_{TFP} = 0.3627 \times z_{EFF} + 1.1214$)

**ii) Aggregate production function (Cobb−Douglas):**

**$Y = \exp(1.1214) \times K_{services}^{0.3156} \times hL^{0.6844} \times EFF^{0.3627}$**

**iii) Useful exergy intensity:**

Approximately constant (1960–2019). Mean X_useful/GDP = 0.5470 (CV=4.7%)

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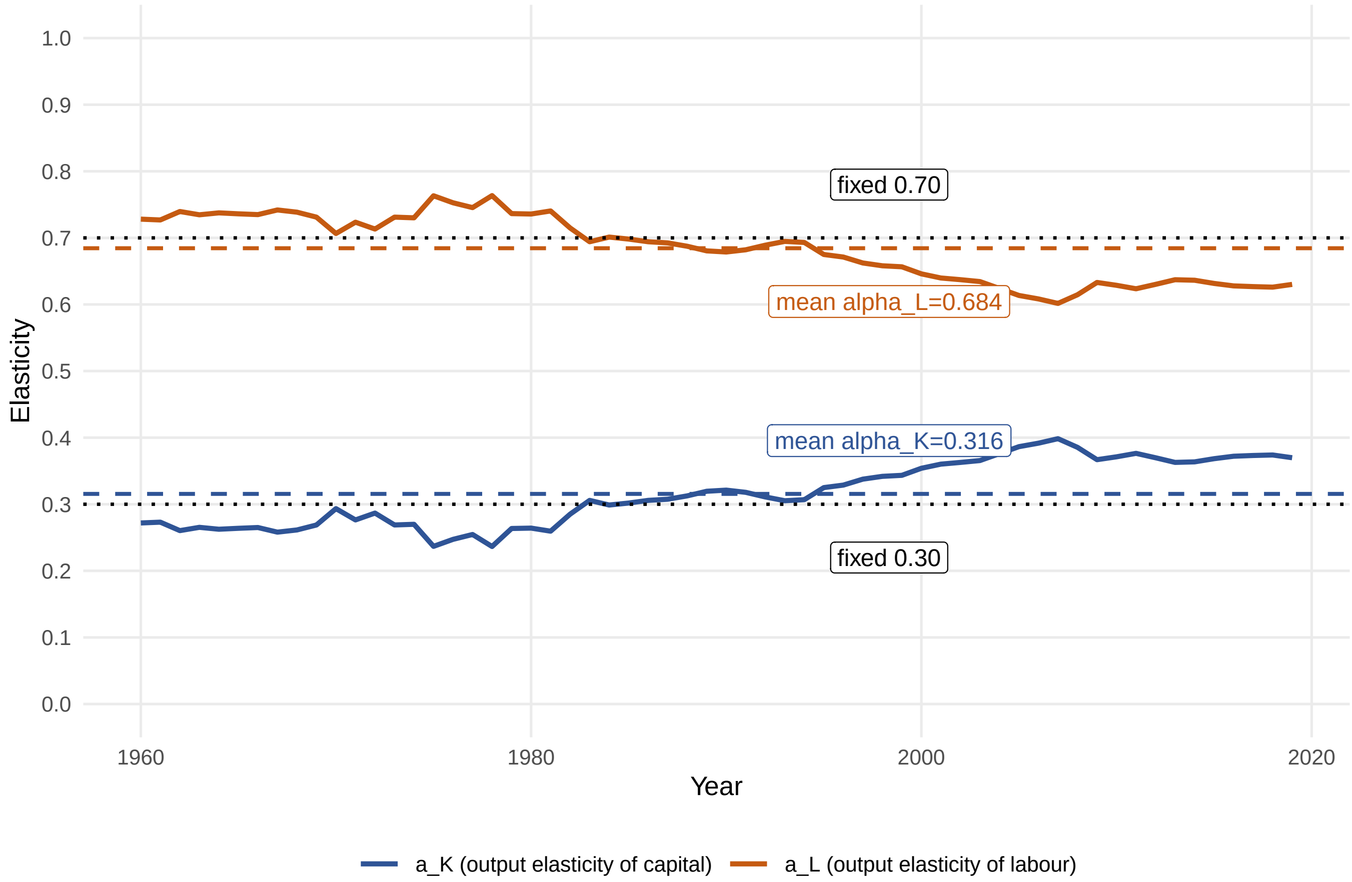
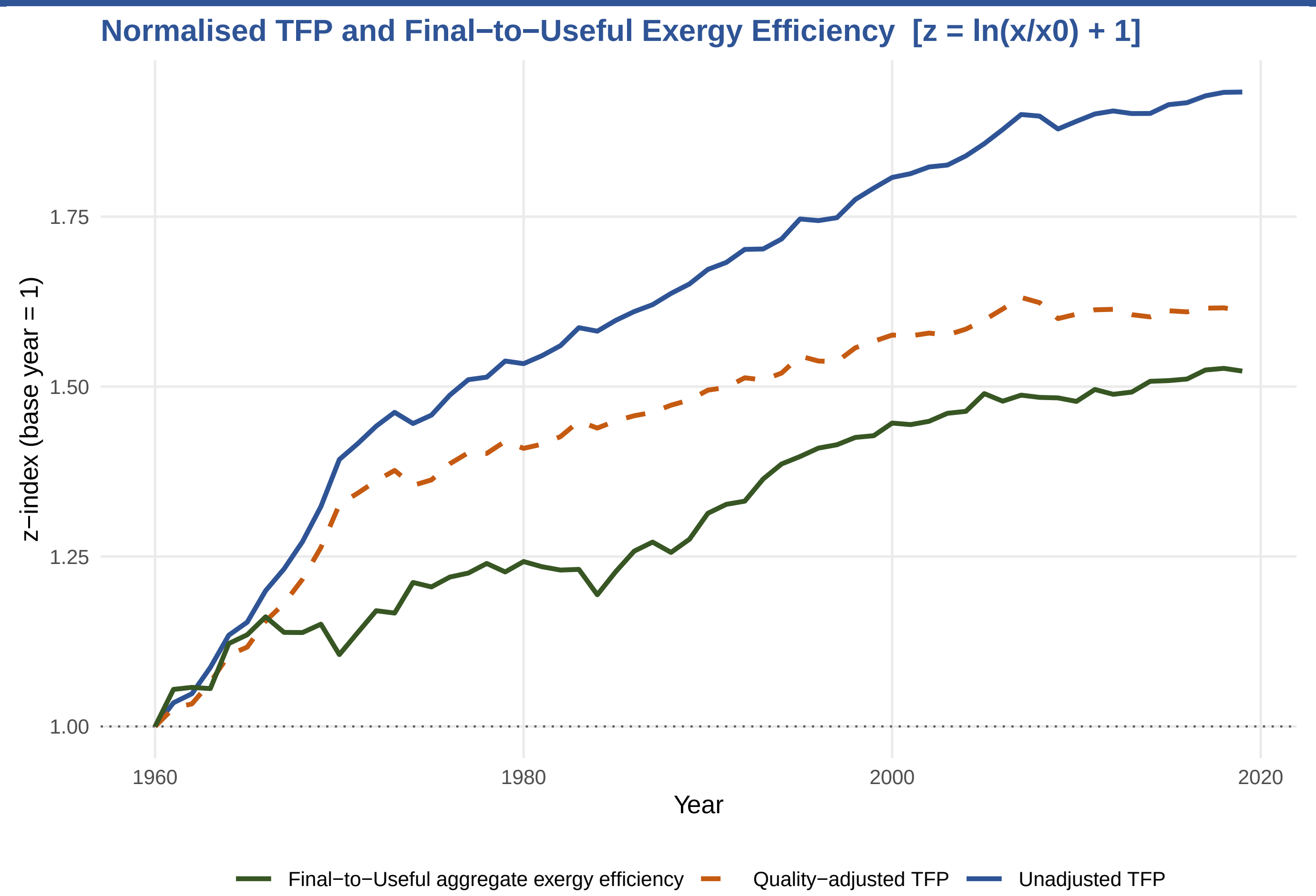
## Annual Output Elasticities




Normalised TFP and Final−to−Useful Exergy Efficiency [z = ln(x/x0) + 1]
z−index (base year = 1)
1.75
1.50
1.25
1.00
1960
1980
2000
2020
Year
Final−to−Useful aggregate exergy efficiency
Quality−adjusted TFP
Unadjusted TFP

## Unit Root Tests — ADF and PP (5% significance level)

**Series: z_tfp_unadj | Verdict: I(1) | ADF: I(1) | PP: I(1)**

| Test | Specification | Statistic | CV 1% | CV 5% | CV 10% | Aux. | Val. | H0 rej.? | Verdict |
|---|---|---|---|---|---|---|---|---|---|
| ADF | Levels (trend+intercept) | −2.4221 | −4.1211 | −3.4425 | −3.1070 | Lags (BIC) | 0 | No | I(1) |
| ADF | 1st diff. (intercept) | −4.7988 | −3.3232 | −2.7807 | −2.5052 | Lags (BIC) | 0 | Yes | — |
| PP | Levels (trend+intercept) | −2.3573 | −4.1190 | −3.4862 | −3.1711 | BW (NW) | 3 | No | I(1) |
| PP | 1st diff. (intercept) | −4.8313 | −3.5456 | −2.9118 | −2.5932 | BW (NW) | 3 | Yes | — |

**Series: z_tfp_adj | Verdict: I(1) | ADF: I(1) | PP: I(1)**

| Test | Specification | Statistic | CV 1% | CV 5% | CV 10% | Aux. | Val. | H0 rej.? | Verdict |
|---|---|---|---|---|---|---|---|---|---|
| ADF | Levels (trend+intercept) | −2.3584 | −4.1211 | −3.4425 | −3.1070 | Lags (BIC) | 0 | No | I(1) |
| ADF | 1st diff. (intercept) | −4.9419 | −3.3232 | −2.7807 | −2.5052 | Lags (BIC) | 0 | Yes | — |
| PP | Levels (trend+intercept) | −2.3078 | −4.1190 | −3.4862 | −3.1711 | BW (NW) | 3 | No | I(1) |
| PP | 1st diff. (intercept) | −4.9907 | −3.5456 | −2.9118 | −2.5932 | BW (NW) | 3 | Yes | — |

**Series: z_eff | Verdict: I(1) | ADF: I(1) | PP: I(1)**

| Test | Specification | Statistic | CV 1% | CV 5% | CV 10% | Aux. | Val. | H0 rej.? | Verdict |
|---|---|---|---|---|---|---|---|---|---|
| ADF | Levels (trend+intercept) | −2.8700 | −4.1211 | −3.4425 | −3.1070 | Lags (BIC) | 0 | No | I(1) |
| ADF | 1st diff. (intercept) | −9.3107 | −3.3232 | −2.7807 | −2.5052 | Lags (BIC) | 0 | Yes | — |
| PP | Levels (trend+intercept) | −2.9504 | −4.1190 | −3.4862 | −3.1711 | BW (NW) | 3 | No | I(1) |
| PP | 1st diff. (intercept) | −9.2961 | −3.5456 | −2.9118 | −2.5932 | BW (NW) | 3 | Yes | — |

**[OK] All series are I(1): unit roots confirmed in levels, stationary in first differences.**

**Green light — Johansen cointegration analysis is appropriate.**

# VAR Residual Portmanteau Tests

## VAR(2): TFP_unadj ~ EFF | T=58 (1962–2019)

| Lags | Q−Stat | Prob. | Adj Q−Stat | Prob. | df |
|---|---|---|---|---|---|
| h=1 | 0.328751 | −−− | 0.334518 | −−− | −−− |
| h=2 | 2.019173 | −−− | 2.085313 | −−− | −−− |
| h=3 | 3.376397 | 0.4969 | 3.516567 | 0.4754 | 4 |
| h=4 | 6.812233 | 0.5570 | 7.206910 | 0.5145 | 8 |
| h=5 | 13.719366 | 0.3190 | 14.765659 | 0.2545 | 12 |

** Test is valid only for lags larger than the VAR lag order.*

## VAR(2): TFP_adj ~ EFF | T=58 (1962–2019)

| Lags | Q−Stat | Prob. | Adj Q−Stat | Prob. | df |
|---|---|---|---|---|---|
| h=1 | 0.371211 | −−− | 0.377724 | −−− | −−− |
| h=2 | 1.764513 | −−− | 1.820787 | −−− | −−− |
| h=3 | 2.875029 | 0.5790 | 2.991876 | 0.5592 | 4 |
| h=4 | 6.773639 | 0.5612 | 7.179272 | 0.5174 | 8 |
| h=5 | 13.691551 | 0.3208 | 14.749816 | 0.2554 | 12 |

** Test is valid only for lags larger than the VAR lag order.*

**[OK] No significant autocorrelation detected (Portmanteau, all valid lags p > 0.05). Green light for cointegration.**

## VAR Residual Serial Correlation (LM) Tests

### VAR(2): TFP_unadj ~ EFF | T=58 (1962–2019)

Null: No serial correlation at lag h

| Lag | LRE* stat | df | Prob. | Rao F−stat | df | Prob. |
|---|---|---|---|---|---|---|
| h=1 | 3.979851 | 4 | 0.4087 | 0.975236 | (4, 100.0) | 0.4246 |
| h=2 | 1.648418 | 4 | 0.8001 | 0.399411 | (4, 100.0) | 0.8087 |
| h=3 | 1.391775 | 4 | 0.8456 | 0.336810 | (4, 100.0) | 0.8526 |
| h=4 | 3.506233 | 4 | 0.4769 | 0.857213 | (4, 100.0) | 0.4925 |
| h=5 | 7.640849 | 4 | 0.1057 | 1.905898 | (4, 100.0) | 0.1153 |

Null: No serial correlation at lags 1 to h

| Lag | LRE* stat | df | Prob. | Rao F−stat | df | Prob. |
|---|---|---|---|---|---|---|
| h=1 | 3.979851 | 4 | 0.4087 | 0.975236 | (4, 100.0) | 0.4246 |
| h=2 | 8.343178 | 8 | 0.4007 | 1.002342 | (8, 96.0) | 0.4395 |
| h=3 | 10.209913 | 12 | 0.5976 | 0.790838 | (12, 92.0) | 0.6586 |
| h=4 | 12.549965 | 16 | 0.7053 | 0.705406 | (16, 88.0) | 0.7814 |
| h=5 | 21.595019 | 20 | 0.3629 | 0.969257 | (20, 84.0) | 0.5062 |

** Edgeworth expansion corrected likelihood ratio statistic.*

### VAR(2): TFP_adj ~ EFF | T=58 (1962–2019)

Null: No serial correlation at lag h

| Lag | LRE* stat | df | Prob. | Rao F−stat | df | Prob. |
|---|---|---|---|---|---|---|
| h=1 | 3.986179 | 4 | 0.4079 | 0.976816 | (4, 100.0) | 0.4238 |
| h=2 | 1.320662 | 4 | 0.8579 | 0.319491 | (4, 100.0) | 0.8643 |
| h=3 | 1.099235 | 4 | 0.8944 | 0.265640 | (4, 100.0) | 0.8994 |
| h=4 | 3.920395 | 4 | 0.4169 | 0.960390 | (4, 100.0) | 0.4328 |
| h=5 | 7.597845 | 4 | 0.1075 | 1.894775 | (4, 100.0) | 0.1172 |

Null: No serial correlation at lags 1 to h

| Lag | LRE* stat | df | Prob. | Rao F−stat | df | Prob. |
|---|---|---|---|---|---|---|
| h=1 | 3.986179 | 4 | 0.4079 | 0.976816 | (4, 100.0) | 0.4238 |
| h=2 | 9.320138 | 8 | 0.3160 | 1.125060 | (8, 96.0) | 0.3536 |
| h=3 | 10.502927 | 12 | 0.5719 | 0.814700 | (12, 92.0) | 0.6348 |
| h=4 | 13.579773 | 16 | 0.6300 | 0.767157 | (16, 88.0) | 0.7174 |
| h=5 | 23.198809 | 20 | 0.2791 | 1.049590 | (20, 84.0) | 0.4169 |

** Edgeworth expansion corrected likelihood ratio statistic.*

**[OK] No significant serial correlation detected (LM, all valid lags p > 0.05). Green light for cointegration.**

## VAR Residual Normality Tests

### VAR(2): TFP_unadj ~ EFF | T=58 (1962–2019)

| Component | Skewness | Chi−sq | df | p | Kurtosis | Chi−sq | df | p | JB stat | df | p |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Eq.1 | −0.166725 | 0.268706 | 1 | 0.6042 | 4.642138 | 6.516827 | 1 | 0.0107 | 6.785533 | 2 | 0.0336 |
| Eq.2 | −0.234886 | 0.533323 | 1 | 0.4652 | 5.172821 | 11.409449 | 1 | 0.0007 | 11.942773 | 2 | 0.0026 |

| Test | Statistic | df | Prob |
|---|---|---|---|
| Joint skewness | 0.802029 | 2 | 0.6696 |
| Joint kurtosis | 17.926276 | 2 | 0.0001 |
| Joint JB | 18.728306 | 4 | 0.0009 |

### VAR(2): TFP_adj ~ EFF | T=58 (1962–2019)

| Component | Skewness | Chi−sq | df | p | Kurtosis | Chi−sq | df | p | JB stat | df | p |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Eq.1 | −0.111625 | 0.120448 | 1 | 0.7285 | 4.654213 | 6.613014 | 1 | 0.0101 | 6.733462 | 2 | 0.0345 |
| Eq.2 | −0.181505 | 0.318459 | 1 | 0.5725 | 4.895694 | 8.684672 | 1 | 0.0032 | 9.003131 | 2 | 0.0111 |

| Test | Statistic | df | Prob |
|---|---|---|---|
| Joint skewness | 0.438907 | 2 | 0.8030 |
| Joint kurtosis | 15.297686 | 2 | 0.0005 |
| Joint JB | 15.736593 | 4 | 0.0034 |

**[!!] WARNING: Non−normal residuals detected (joint JB p < 0.05). Cointegration p−values may be affected.**

# Johansen Cointegration Tests: Hz (no deterministic term)

## TFP_unadj ~ EFF | Hz (no deterministic term) | Lags (1st diff.): 1 to 2 | T=58 (1962–2019)

### Unrestricted Cointegration Rank Test (Trace)

| Hypothesized | Eigenvalue | Statistic | CV 5% | Prob.** |
|---|---|---|---|---|
| None (r=0) * | 0.339791 | 26.05431 | 17.95000 | 0.0044 |
| At most 1 (r=1) | 0.041029 | 2.38797 | 8.18000 | 0.5346 |

* Trace indicates 1 CE at 5%

** MacKinnon−Haug−Michelis (1999) p−values.

### Unrestricted Cointegration Rank Test (Max−Eigenvalue)

| Hypothesized | Eigenvalue | Statistic | CV 5% | Prob.** |
|---|---|---|---|---|
| None (r=0) * | 0.339791 | 23.66635 | 14.90000 | 0.0016 |
| At most 1 (r=1) | 0.041029 | 2.38797 | 8.18000 | 0.5346 |

* Max−Eigenvalue indicates 1 CE at 5%

** MacKinnon−Haug−Michelis (1999) p−values.

### Normalised Cointegrating Coefficients (std. error in parentheses):

| Variable | Coefficient | Std. Error |
|---|---|---|
| TFP_unadj | 1.000000 | |
| EFF | −0.958553 | (0.02002) |

**beta=0.9586 > 0 Valid**

## TFP_adj ~ EFF | Hz (no deterministic term) | Lags (1st diff.): 1 to 2 | T=58 (1962–2019)

### Unrestricted Cointegration Rank Test (Trace)

| Hypothesized | Eigenvalue | Statistic | CV 5% | Prob.** |
|---|---|---|---|---|
| None (r=0) * | 0.337106 | 25.27909 | 17.95000 | 0.0057 |
| At most 1 (r=1) | 0.031834 | 1.84407 | 8.18000 | 0.6445 |

* Trace indicates 1 CE at 5%

** MacKinnon−Haug−Michelis (1999) p−values.

### Unrestricted Cointegration Rank Test (Max−Eigenvalue)

| Hypothesized | Eigenvalue | Statistic | CV 5% | Prob.** |
|---|---|---|---|---|
| None (r=0) * | 0.337106 | 23.43502 | 14.90000 | 0.0018 |
| At most 1 (r=1) | 0.031834 | 1.84407 | 8.18000 | 0.6445 |

* Max−Eigenvalue indicates 1 CE at 5%

** MacKinnon−Haug−Michelis (1999) p−values.

### Normalised Cointegrating Coefficients (std. error in parentheses):

| Variable | Coefficient | Std. Error |
|---|---|---|
| TFP_adj | 1.000000 | |
| EFF | −0.415953 | (0.01966) |

**beta=0.4160 > 0 Valid**

## Johansen Cointegration Tests: Hc (restricted constant; no trend)

### TFP_unadj ~ EFF | Hc (restricted constant; no trend) | Lags (1st diff.): 1 to 2 | T=58 (1962–2019)

#### Unrestricted Cointegration Rank Test (Trace)

| Hypothesized | Eigenvalue | Statistic | CV 5% | Prob.** |
|---|---|---|---|---|
| None (r=0) * | 0.396051 | 33.87160 | 19.96000 | 0.0002 |
| At most 1 (r=1) | 0.086044 | 5.12844 | 9.24000 | 0.2544 |

* Trace indicates 1 CE at 5%

** MacKinnon−Haug−Michelis (1999) p−values.

#### Unrestricted Cointegration Rank Test (Max−Eigenvalue)

| Hypothesized | Eigenvalue | Statistic | CV 5% | Prob.** |
|---|---|---|---|---|
| None (r=0) * | 0.396051 | 28.74316 | 15.67000 | 0.0003 |
| At most 1 (r=1) | 0.086044 | 5.12844 | 9.24000 | 0.2544 |

* Max−Eigenvalue indicates 1 CE at 5%

** MacKinnon−Haug−Michelis (1999) p−values.

#### Normalised Cointegrating Coefficients (std. error in parentheses):

| Variable | Coefficient | Std. Error |
|---|---|---|
| TFP_unadj | 1.000000 | |
| EFF | −0.905248 | (0.16543) |
| C | −0.639561 | (0.23994) |

**beta=0.9052 > 0 Valid**

### TFP_adj ~ EFF | Hc (restricted constant; no trend) | Lags (1st diff.): 1 to 2 | T=58 (1962–2019)

#### Unrestricted Cointegration Rank Test (Trace)

| Hypothesized | Eigenvalue | Statistic | CV 5% | Prob.** |
|---|---|---|---|---|
| None (r=0) * | 0.388275 | 33.90945 | 19.96000 | 0.0002 |
| At most 1 (r=1) | 0.098262 | 5.89557 | 9.24000 | 0.1905 |

* Trace indicates 1 CE at 5%

** MacKinnon−Haug−Michelis (1999) p−values.

#### Unrestricted Cointegration Rank Test (Max−Eigenvalue)

| Hypothesized | Eigenvalue | Statistic | CV 5% | Prob.** |
|---|---|---|---|---|
| None (r=0) * | 0.388275 | 28.01389 | 15.67000 | 0.0004 |
| At most 1 (r=1) | 0.098262 | 5.89557 | 9.24000 | 0.1905 |

* Max−Eigenvalue indicates 1 CE at 5%

** MacKinnon−Haug−Michelis (1999) p−values.

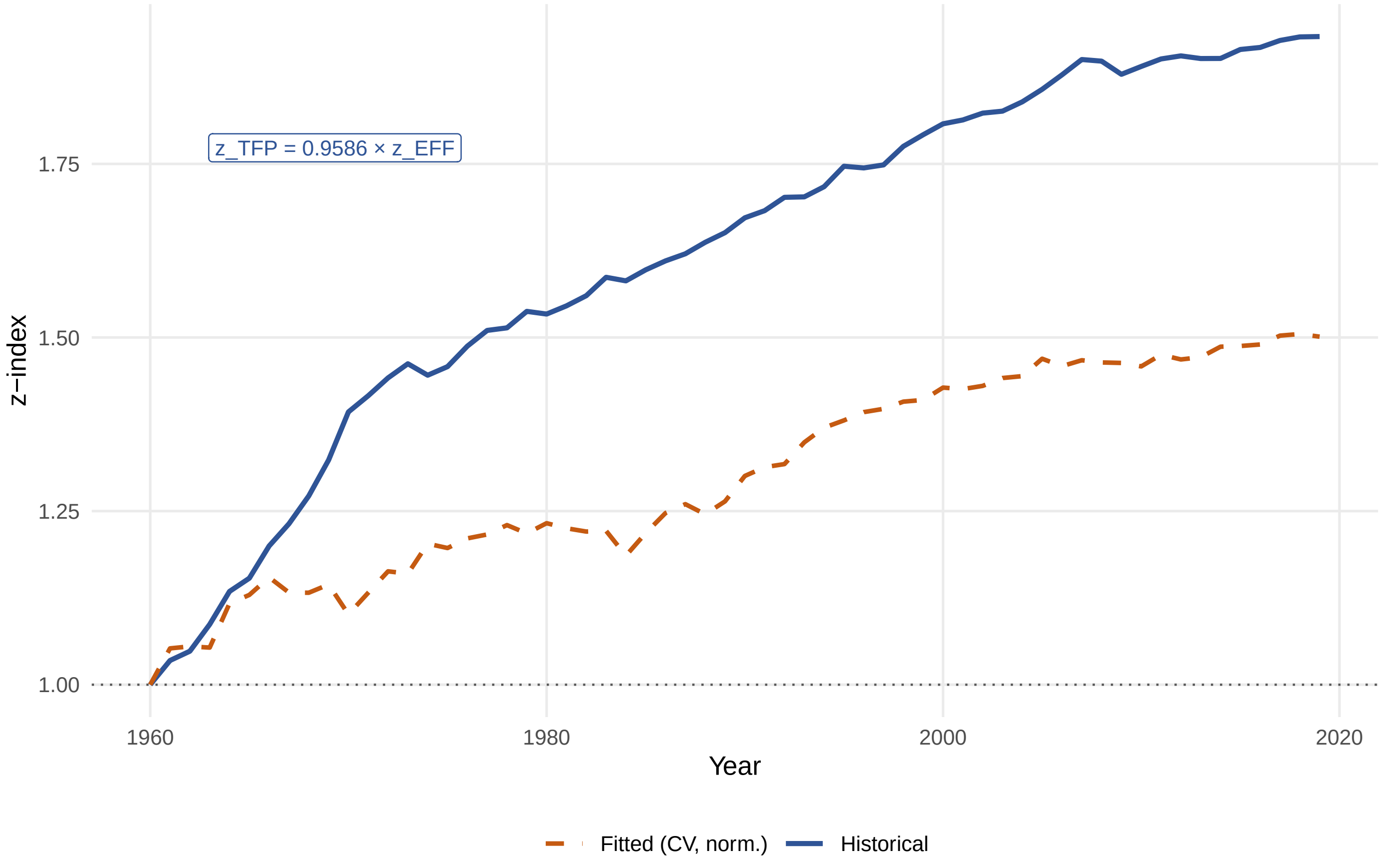
#### Normalised Cointegrating Coefficients (std. error in parentheses):

| Variable | Coefficient | Std. Error |
|---|---|---|
| TFP_adj | 1.000000 | |
| EFF | −0.362679 | (0.16530) |
| C | −1.121372 | (0.23370) |

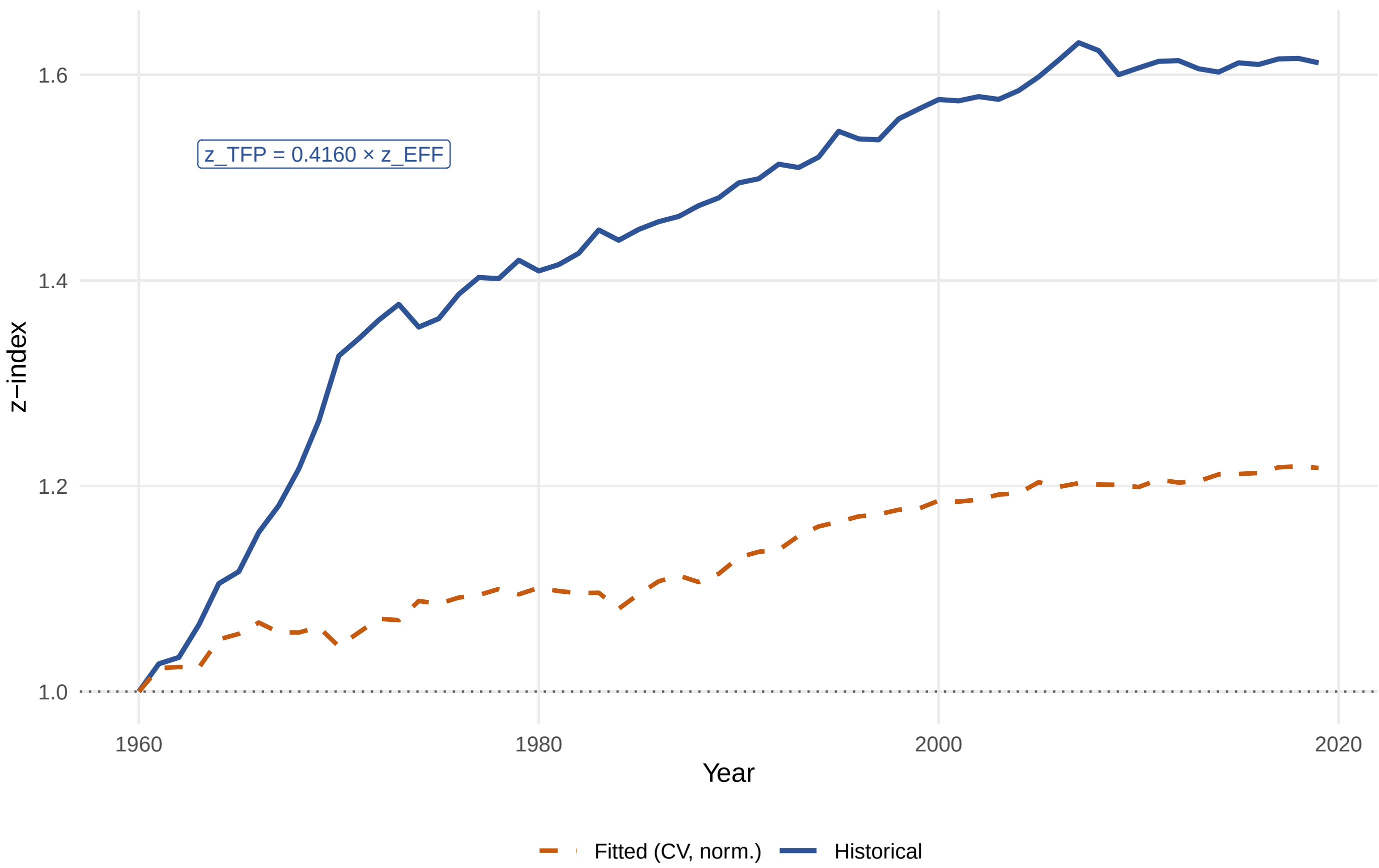
**beta=0.3627 > 0 Valid**

## Historical vs Fitted TFP [Hz (no deterministic term)]

### Unadjusted TFP

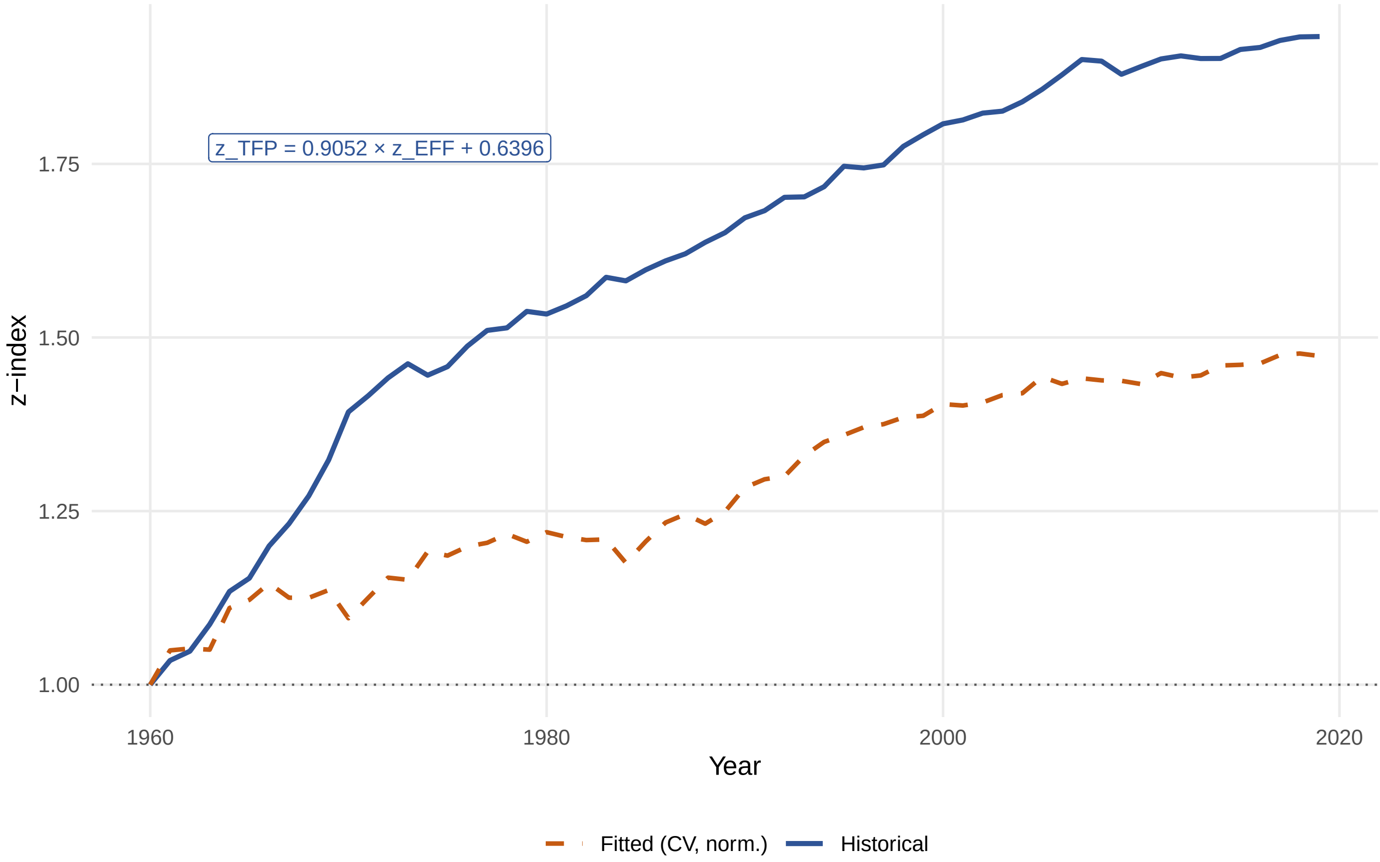



### Quality−adjusted TFP

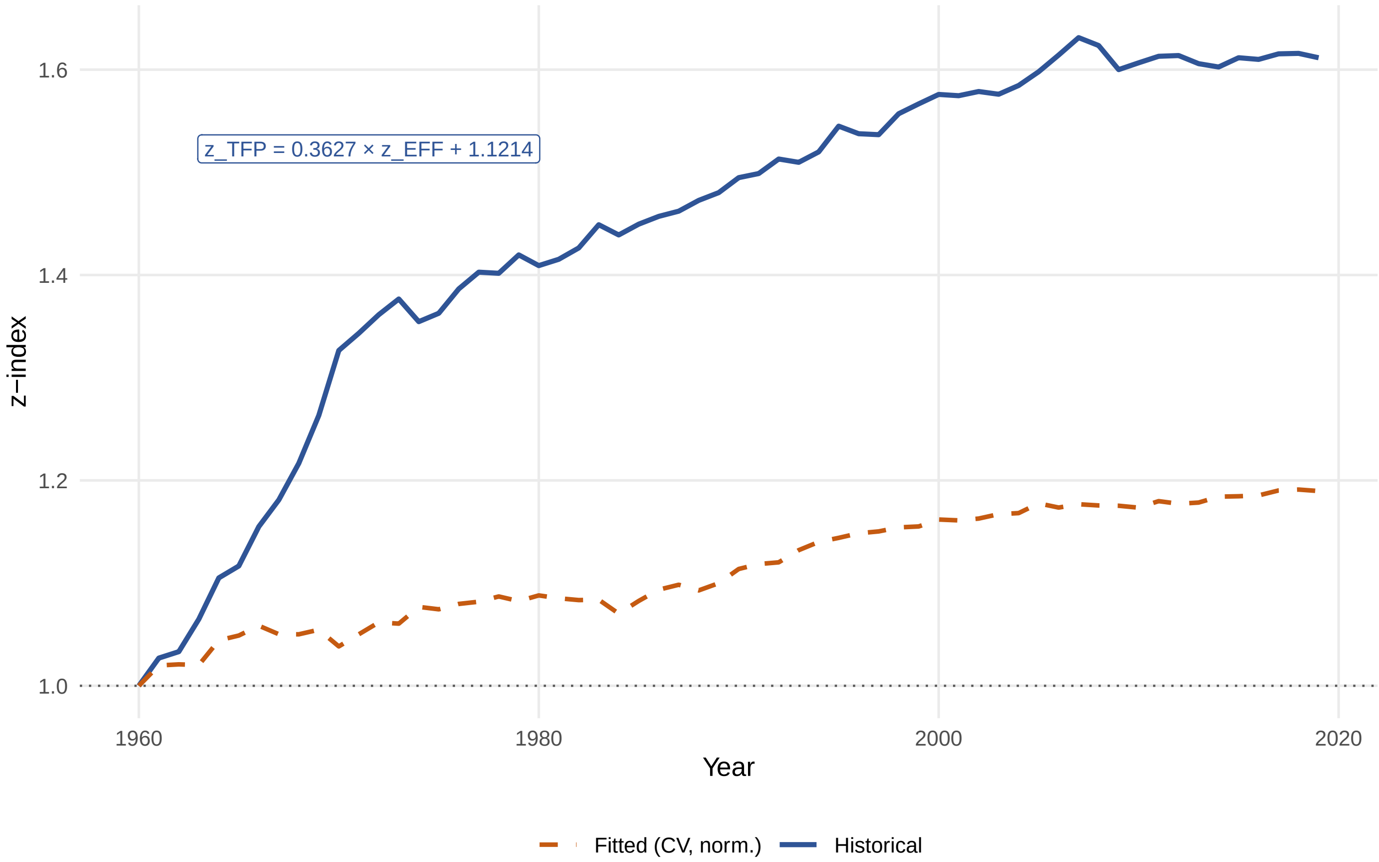


## Historical vs Fitted TFP [Hc (restricted constant; no trend)]

### Unadjusted TFP

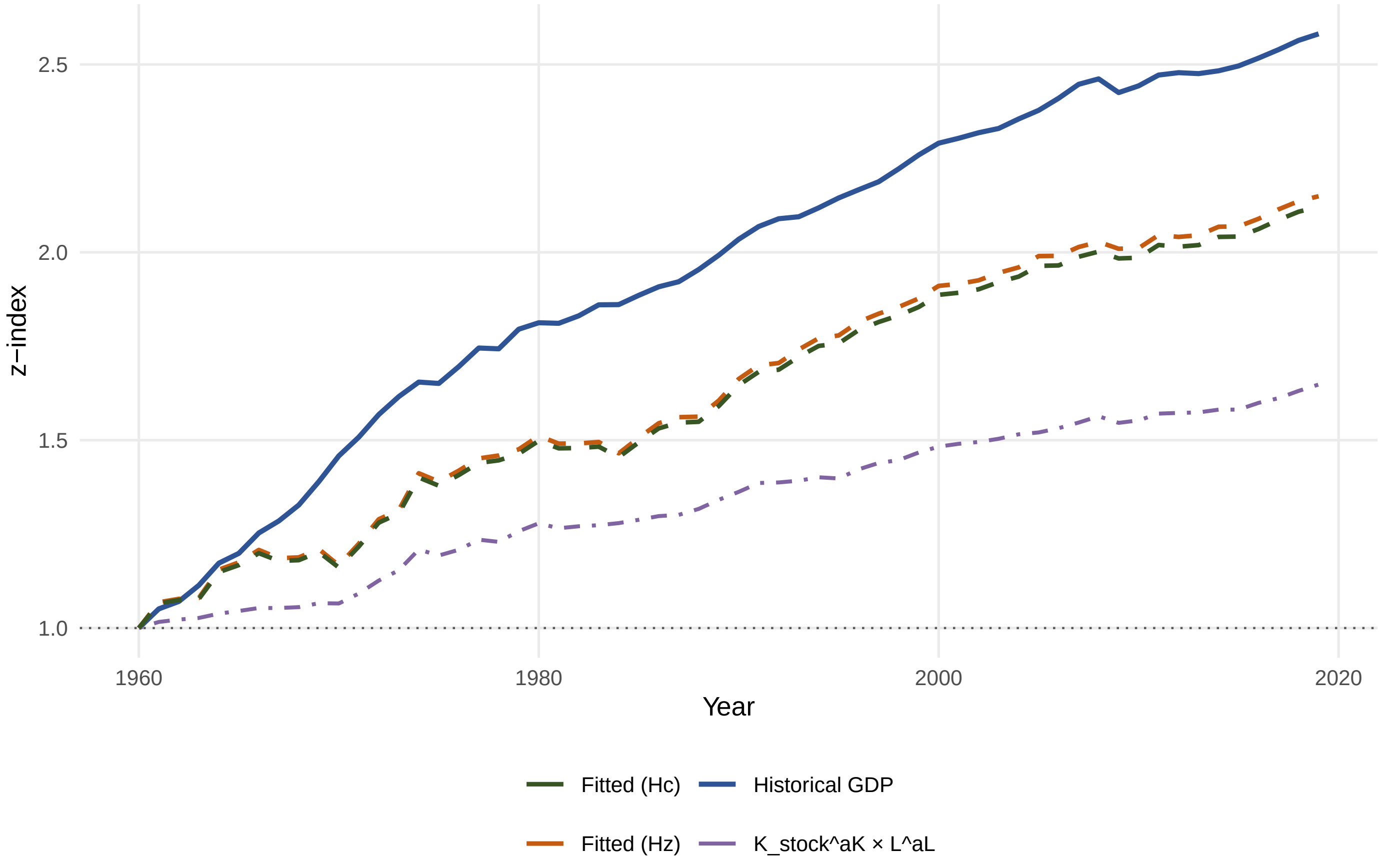



### Quality−adjusted TFP

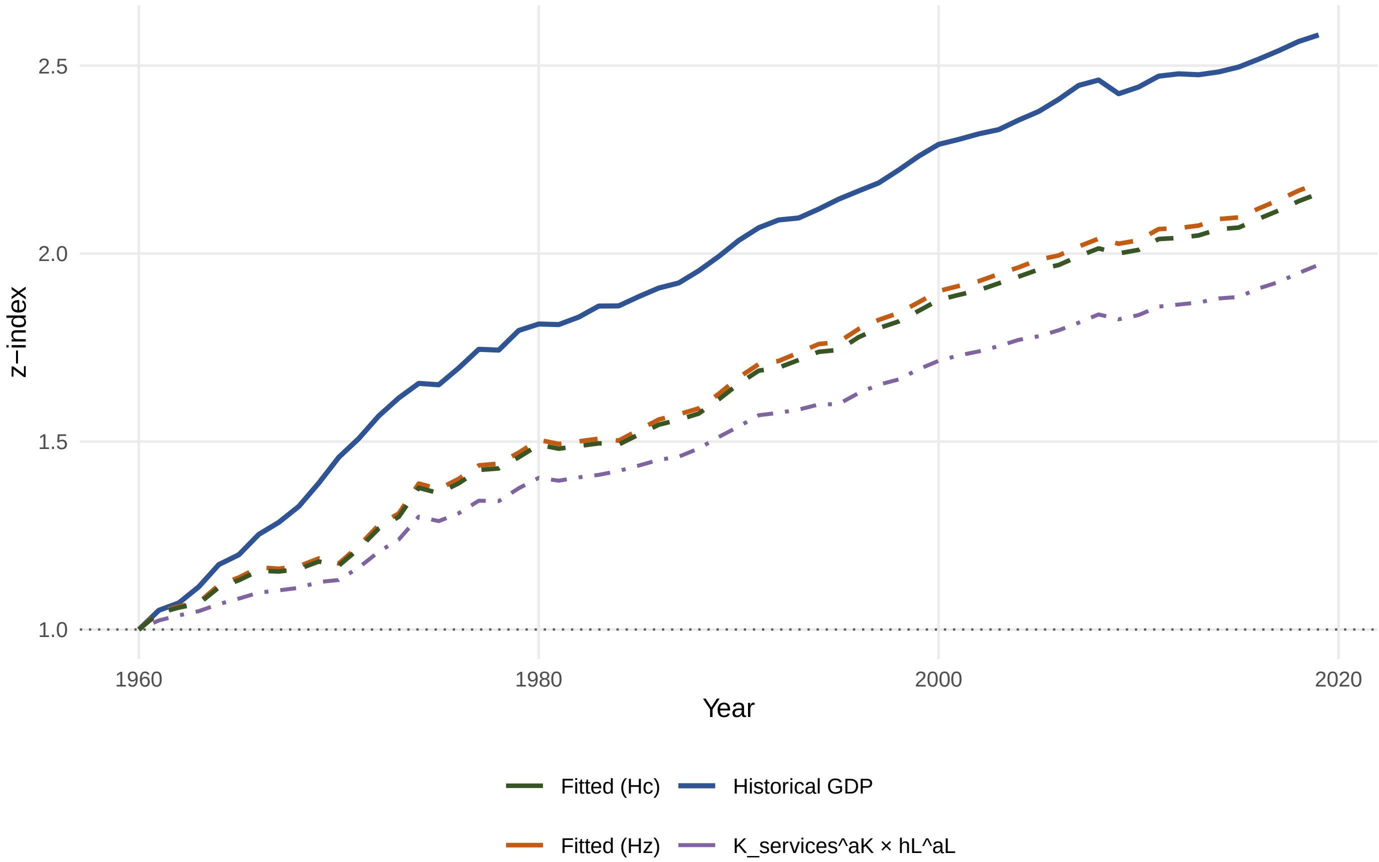


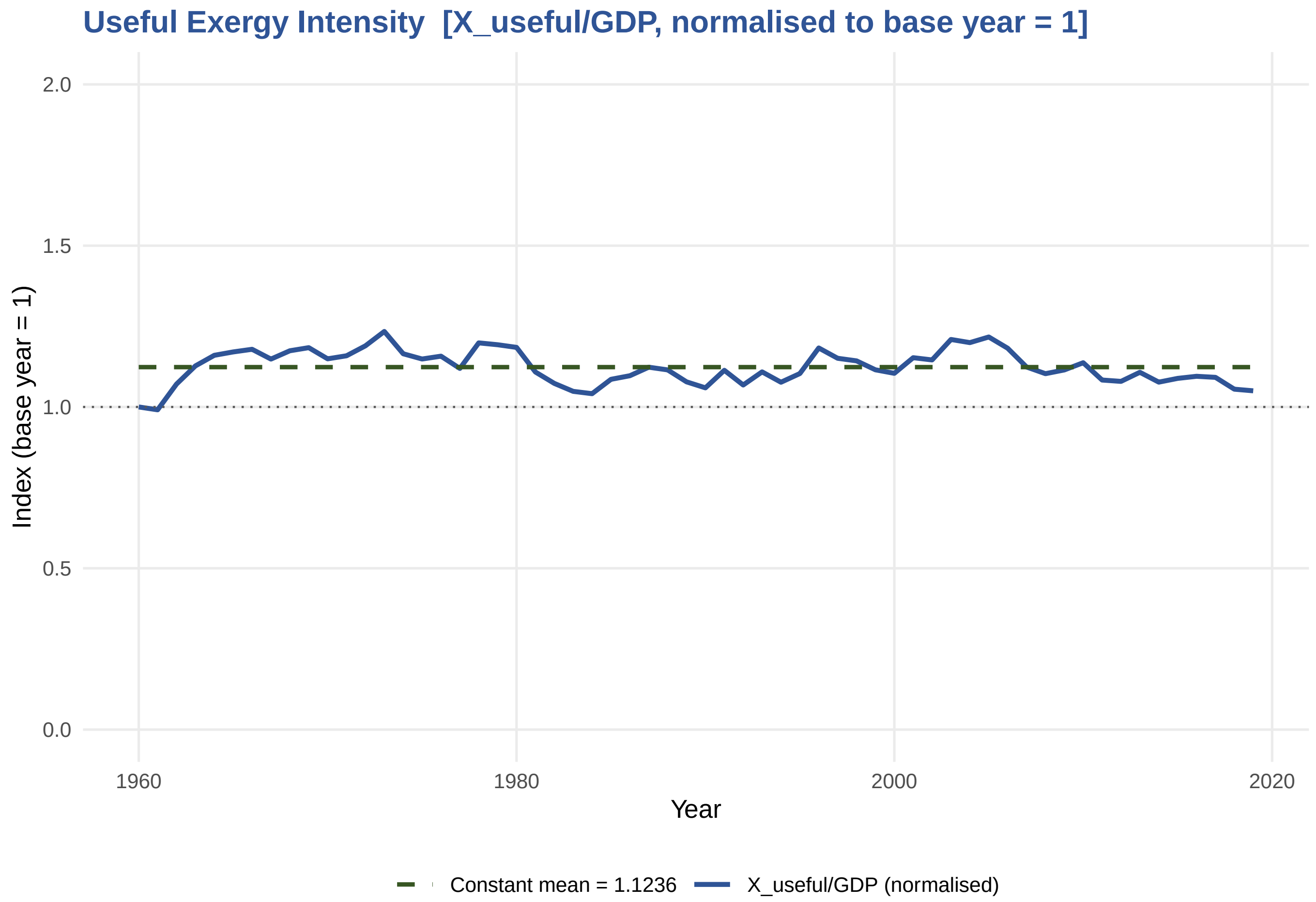
Historical vs Fitted GDP [both Hz and Hc specifications]

## Unadjusted (K_stock, L)




## Quality−adjusted (K_services, hL)






| Statistic | Full (1960–2019, n=60) | Sub (2000–2019, n=20) |
|---|---|---|
| Mean intensity | 0.5470 | 0.5457 |
| Std deviation | 0.0258 | 0.0241 |
| CV (%) | 4.72 | 4.42 |
| Min | 0.4826 | 0.5113 |
| Max | 0.6008 | 0.5925 |
| OLS slope (per year) | −0.000231 | −0.003018 |
| Trend t−stat | −1.205 | −4.674 |
| Trend p−value | 0.2329 | 0.0002 |
| Trend R2 | 0.0244 | 0.5483 |
| Annual growth rate (% p.a.) | −0.042 | −0.553 |
| ADF statistic (levels) | −3.4651 | −3.2757 |
| ADF CV 5% | −3.4425 | −3.1875 |
| PP statistic (levels) | −3.4555 | −3.2613 |
| PP CV 5% | −3.4862 | −3.6746 |

### Individual test results

| Statistic | Full (1960–2019, n=60) | Sub (2000–2019, n=20) |
|---|---|---|
| No significant trend (p>0.10) | Yes | No |
| ADF: I(0) at 5% | Yes | Yes |
| PP: I(0) at 5% | No | No |
| CV < 5% | Yes | Yes |
| VERDICT: Approx. constant? | Yes | Yes |

## References

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